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# BiCrossNet with Decoupled Dual Generators: A Parameter-Efficient and Generalizable Few-Shot Custom Gesture Recognition Framework

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## Abstract

Gesture interaction and touchless sensing is a natural and intuitive control method that allows users to control devices through natural hand or body movements, reducing reliance on physical input and enhancing convenience and functional efficiency. The biggest challenge is balancing accuracy and model parameter count for custom gesture tasks, while minimizing the number of data samples required for training. This paper proposes a novel IMU-based BiCrossNet model and two novel data augmentation models (delta-generator and embedding-generator) to address this challenge. Compared with existing methods, the model proposed in this article boosts accuracy by 11.7% and 12.7% in UMAHand (public datasets); and 8.85% and 5.25% in GRHand (self-developed datasets), with decreasing 27.8% pre-trained feature-extractor model parameters. This research lays a solid foundation for deploying and implementing custom gesture recognition engineering on intelligent terminal devices.

## 1. Introduction

Gesture recognition, especially custom gesture recognition offers richer input methods and enables users to interact with smart wearable devices (such as smart watch, smart bracelet, XR glass, etc.) using their respective habits in more flexible and diverse ways (Xu et al., 2015). According to MarketsandMarkets™, the global Gesture Recognition and Touchless Sensing Market is projected to reach USD 125.9 billion by 2032 from USD 28.4 billion in 2025, at a CAGR of 23.7%.

Researchers and experts from universities and high-tech companies have been deeply involved in this field for many

years, and a variety of technologies have been developed, refined, and deployed to facilitate rapid deployment and commercialization (Choi et al., 2025; Gomes et al., 2023). A hand gesture recognition algorithm based on wearable photoplethysmography (PPG) sensors integrated with green, red, and infrared light sources is proposed to achieve favorable recognition accuracy even with a relatively small volume of training data (Li et al., 2023). However, PPG-based algorithms are inherently plagued by several drawbacks, primarily arising from sensor constraints, environmental interference, physiological individual variability, and algorithmic limitations. To tackle the few-shot learning challenge in gesture recognition, a dedicated hand gesture recognition approach based on monocular cameras is presented, which leverages transformer architectures and meta-learning techniques (Shahi et al., 2024). Nevertheless, this algorithm remains highly susceptible to variations in illumination conditions and complex ambient backgrounds, while also suffering from a trade-off between real-time inference performance and computational resource. To alleviate the vulnerability of conventional sensors to environmental factors, Apple Inc proposed a hand gesture customization method based on inertial measurement unit (IMU) sensors and devised a customizable gesture detection framework that requires only minimal annotated data, and further put forward a few-shot learning paradigm that distills lightweight models from pre-trained networks (Xu et al., 2022), thereby enabling lossless knowledge transfer, the drawbacks of this method are that the pretrain\_workflow model has a large number of parameters and poor accuracy; the memory usage of the data augmentation module in the customization\_workflow is linearly related to the number of delta vectors, and the model trained with the data augmented using it has poor generalization ability.

As outlined in the above introduction, custom gesture recognition technology can bring a better human-computer interaction experience, and various sensors and technologies can be used for gesture recognition. However, the commercialization of this technology still faces many challenges:

- Small model footprint and parameter size are needed as excessive ones hinder deployment on resource-constrained smart wearable devices.

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- Good generalization of the model is needed as poor generalization demands massive motion data for new gesture development and causes high costs of large-scale data collection.
- High accuracy of the algorithm is needed as low accuracy severely impacts the user experience.
- Low power consumption is needed as high power consumption impairs device battery life.

Therefore, the main objective of this paper is to propose and implement a few-sample custom gesture recognition framework and algorithm to effectively overcome or mitigate the challenges faced in the commercialization of this technology. The main work of this research can be summarized as:

- A dual-branch single-modal feature extraction module and a symmetric cross-attention module are proposed in BiCrossNet to address the issues of large parameter counts and poor performance in pre-trained models, enhance feature discriminative power, avoids redundant intra-modal computation, and provide excellent interpretability.
- Two novel data augmentation models decouple the memory usage of the data augmentation module from the sample size and optimize the generalization of augmented samples. During inference, only the model needs to be stored, eliminating the need for additional storage of delta vector sets, thus saving memory resources.
- A dataset of more than 40 hours was constructed to test and validate the effectiveness of the proposed model framework and algorithm.

The proposed BiCrossNet framework and models have undergone comprehensive testing and validation on both public dataset and self-developed dataset. In comparison to existing technologies, it demonstrates superior generalization capabilities and is capable of adapting to diverse new gestures.

**Conflict of Interest Disclosure:** The BiCrossNet framework proposed in this paper was developed at OPPO Research Institute, and the GRHand dataset used in this work was funded and constructed by OPPO Research Institute. All authors are employed by OPPO Research Institute.

## 2. Method

This section delineates the architecture of our proposed BiCrossNet framework. For a practical custom gesture recog-

nition system, the algorithm needs to meet two critical requirements: (a) Sample efficiency for rapid adaptation: For supporting new gestures, the training data demand of new gestures must be minimized. Excessive real-world user data collection would significantly degrade usability and adoption rates of the function. (b) On-device efficiency: The model must be sufficiently lightweight for deployment and real-time inference on resource-constrained mobile hardware, failing which practical integration is unfeasible.

To meet the first requirement, the algorithm necessitates a powerful data augmentation mechanism capable of generating abundant, high-quality synthetic samples from a limited real sample set. This enables the model to effectively adapt to novel gesture types. For the second requirement, a lightweight yet robust feature extractor and classifier must be designed. Drawing on extensive experience with mobile interactive perception systems, we target a model that maintains more than 95% average accuracy while constraining the parameter count to approximately 50k, with lower being preferable.

### 2.1. Framework

The **BiCrossNet framework** is presented and depicted in Fig.1 and Fig.2, and it comprises two core components: (a) A high-quality data augmentation module, which consists of a Delta Generator and an Embedding Generator. This module can synthesize a substantial set of high-quality training samples from a very small number of real samples (less than or equal to 5), enabling effective model adaptation to newly added custom gestures. (b) A lightweight yet highly accurate feature extraction network containing only approximately 32k parameters. This design achieves a significant reduction in model size while maintaining and in some cases surpassing the recognition accuracy of larger feature extraction networks. The following sections elaborate on the design of each module.

### 2.2. Delta Generator

Apple Inc.(Xu et al., 2022) introduced a data augmentation model named Delta-Encoder, which is based on a classical encoder-decoder architecture. The model operates on pairs of real samples from the same class. First, a feature extractor processes these samples into feature embeddings. These embeddings are then concatenated and fed into the encoder to produce a delta vector, which represents the difference in feature information between embeddings of different samples within the same class. Subsequently, this delta vector is concatenated with one of the original embeddings and input to the decoder, which generates an augmented sample embedding. The training objective involves maximizing the similarity between this augmented embedding and the other embedding from the original sample pair. Consequently,

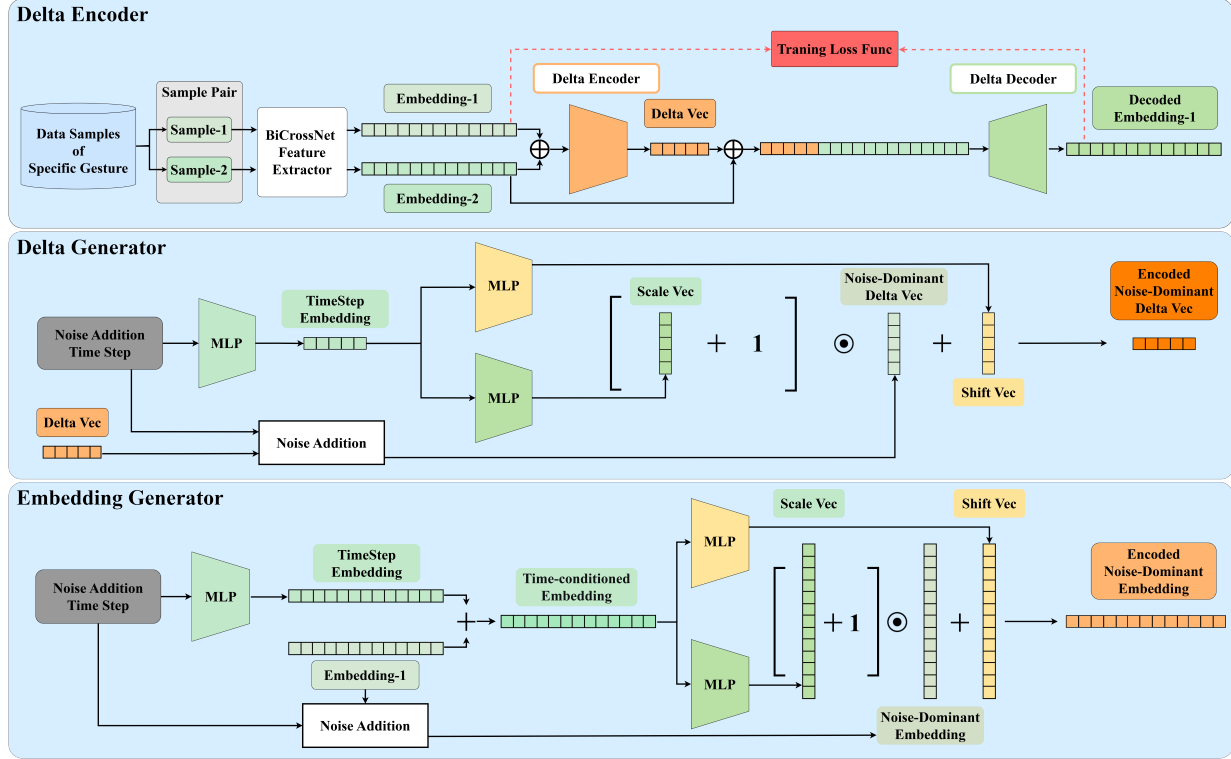


Figure 1. Comparison of Workflows between Our Proposed Data Augmentation Methods (Delta Generator and Embedding Generator) and the Baseline Method (Delta Encoder).

an encoder pre-trained on a large-scale dataset can later, when new custom gestures are introduced, generate delta vectors by sampling pairs from the pre-training data. By combining these vectors with the few available real-sample embeddings from the new class, the decoder can produce a substantial set of high-quality augmented embeddings to support effective adaptation.

Although AppleDE demonstrates promising performance (Xu et al., 2022), it has two significant limitations: (a) Storage inefficiency and poor deployability: The method necessitates the retention of either the complete large-scale pre-training dataset (for dynamic delta-vector sampling) or a vast collection of pre-generated delta vectors. This imposes considerable storage overhead and undermines practical usability in constrained environments. (b) Limited generalization to new gestures: The delta vectors are optimized to capture fine-grained differences between samples of known actions from the pre-training set. Consequently, when synthesizing augmented samples for a novel gesture, the process remains excessively tied to the prior data distribution, which ultimately restricts the model’s ability to generalize effectively.

To address these limitations, we first designed a Delta Generator based on a generative model. Its core architecture follows the diffusion process, comprising a forward noising

stage and a reverse denoising stage. During the forward process, Laplacian noise is iteratively added to a Delta vector until it approaches pure noise (Dwork et al., 2006).

$$\delta_t = \sqrt{\alpha_t} \cdot \delta_0 + \sqrt{1 - \alpha_t} \cdot \varepsilon \quad (1)$$

$$\varepsilon \sim \text{Laplace}(0, b_t) \quad (2)$$

Here, we replace the conventional Gaussian noise with Laplacian noise. This choice is motivated by the Laplacian distribution’s characteristic sharp peak and heavy tails, which offer a higher concentration of low-amplitude perturbations and a lower probability of extreme outliers (Laplace, 1774). This property enables a closer match to the typical structure of Delta vectors, thereby more effectively capturing the subtle variations characteristic of fine-grained motion differences.

The reverse process learns to iteratively denoise the signal by estimating and removing the noise at each step. Through this procedure, the model effectively learns the underlying data distribution of the Delta vectors.

$$\delta_{\text{denoised}} = \frac{\delta_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \cdot \hat{\varepsilon}_\theta(\delta_t, t)}{\sqrt{\alpha_t}} + \sigma_t \cdot \varepsilon' \quad (3)$$

$$\varepsilon' \sim \text{Laplace}(0, b_t) \quad (4)$$

$$\mathcal{L} = \mathbb{E} [\|\varepsilon - \hat{\varepsilon}_\theta(\delta_t, t)\|_1] \quad (5)$$

Here, an L1 loss for training is employed, as it imposes a linear penalty on errors, unlike the MSE loss, which squares and thus amplifies outliers. The L1 loss corresponds to learning the median, which is inherently robust to outliers. This property prevents the model from overfitting to low-probability extreme perturbations at the expense of accurately capturing the core subtle differences. Consequently, the denoised delta vectors remain stable and faithful to the original data distribution.

In summary, the combined use of Laplacian noise and L1 loss prevents the model from overfitting to extreme perturbations. This is achieved by mitigating their influence both at the level of the prior noise distribution and through the training objective. This design ensures the denoising model robustly learns the underlying distribution of delta vectors. Consequently, a pre-trained Delta Generator can synthesize new, target-like samples directly from random noise via the denoising process, eliminating the need to store large-scale pre-training datasets or pre-computed delta vectors. More importantly, during the adaptation to new gesture types, it can generate high-fidelity and diverse synthetic delta vectors to effectively support few-shot learning.

### 2.3. Embedding Generator

Although the Delta Generator exhibits robust performance, notably enhancing the augmentation of novel gesture data and removing the reliance on large-scale pre-trained datasets and a distinct Encoder, its entire workflow still necessitates two pre-trained models: the delta generator itself and a decoder. This configuration does not achieve fully convenient, end-to-end augmentation. To overcome this limitation, we introduce an enhanced Embedding Generator. Its core advancement is the ability to synthesize rich, diverse, and realistic sample embeddings directly from random noise, using only a few real-sample embeddings from the new gesture class as guidance. This design decouples the augmentation process from a dedicated decoder. Through end-to-end training, the model is able to learn how to capture the latent distribution of intra-class feature variations within the embedding space.

### 2.4. BiCrossNet Feature Extractor

Both AppleDE and Improved TCNN (Kim et al., 2024) (The original paper on the Improved TCNN method did not propose corresponding training data augmentation techniques.) incorporate feature extraction models specifically designed for custom gesture recognition, containing 56,752 and 45,198 parameters, respectively. Although these models deliver satisfactory accuracy, our extensive experience in deploying interactive perception algorithms on mobile devices suggests significant potential for further parameter reduction in this task, without compromising performance.

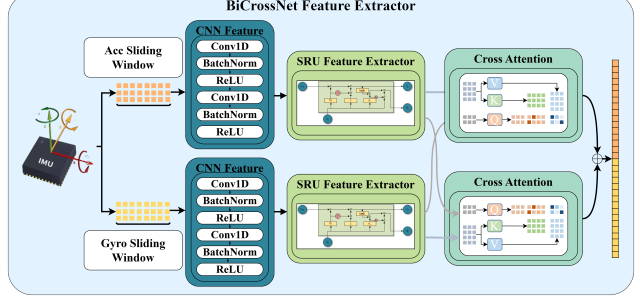


Figure 2. BiCrossNet Feature Extractor Network Architecture.

Motivated by this, we design a dual-branch, cross-attention-driven temporal feature extraction network, as shown in Fig.2.

First, we construct a dual-branch, single-modality feature extraction module that combines CNN and SRU layers. The CNN branch captures local spatial patterns within the IMU gesture signal, while the SRU branch models its temporal dependencies. The two branches process linear acceleration and angular velocity data separately, allowing for a purer and more focused extraction of the inherent spatial structures unique to each physical signal.

Subsequently, we introduce a symmetric cross-attention module. Instead of simply concatenating or adding the features from the two branches, this module enables bidirectional dynamic feature interaction. It allows the acceleration features to query the angular velocity features, and vice versa. This design adaptively captures complementary and synergistic relationships between the two modalities, effectively models inter-modal interactions, avoids redundant intra-modal computations, and ultimately enhances the discriminative power of the learned features.

## 3. Experiments

### 3.1. Dataset

To rigorously evaluate the proposed method and benchmark it against state-of-the-art (SOTA) approaches, we conduct experiments on both a public benchmark dataset UMAHand and a self-developed dataset GRHand, which is recorded using commercial smartwatches in unconstrained real-world scenarios, making it more representative of practical deployment conditions. This dual-dataset strategy allows us to assess general performance on a controlled benchmark and, crucially, practical efficacy in a realistic, few-shot custom gesture scenario. The datasets are described in Table.1.

**GRHand Dataset:** To evaluate performance under practical conditions, we introduce GRHand, a self-developed inertial dataset for gesture interaction in mobile smart device contexts, collected using commercial smartwatches. GRHand comprises two distinct subsets:

Table 1. Introduction of Public UMAHand Dataset and the few-shot custom gesture subset of self-developed GRHand Dataset.

|                           | UMAHand (PUBLIC)                                                                                                 | GRHand (SELF-DEVELOPED)                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PLATFORM                  | a mote or Shimmer sensor device                                                                                  | commercial smart watch                                                                                                                                                 |
| IMU SENSORS               | triaxial acc, gyro, mag, and baro, 100Hz                                                                         | ST LSM6DSOW 6 axes, acc and gyro, 100Hz                                                                                                                                |
| PARTICIPANTS & ACTIVITIES | 25 participants, 29 manual activities (7 used)                                                                   | 10 participants, 9 manual activities (9 used)                                                                                                                          |
| DATA DURATION             | 52 min                                                                                                           | 113 min                                                                                                                                                                |
| GESTURE CATEGORY          | brushing teeth, clapping hands, combing hair, waving hands, buttoning buttons, zipping zippers, screwing screws. | five-finger bend, clapping, wrist down flip, side wrist flip, five finger snap, peace sign with one fist, flip coin, double wrist flip, snapping.                      |
| DEVICE & SCENARIOS        | laboratory-grade IMU collects data in a controlled home environment, ensuring the data is pure and standardized. | commercial smartwatches collect data in unconstrained real-world scenarios, directly resolving the distribution offset issue from laboratory to real-world deployment. |
| GESTURE DEFINITION LOGIC  | definitions are biased towards continuous daily activities, assessing "activity awareness" capabilities.         | definitions focus on transient interactive commands, assessing "interaction understanding" capabilities.                                                               |

Table 2. Detail Information about the large-scale pre-training subset of self-developed GRHand Dataset.

|                          | GRHand (SELF-DEVELOPED) |                                                                                                                 |            |
|--------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------|------------|
|                          | Training                | Test                                                                                                            | Validation |
| Number OF DATA SEQUENCES | 100619                  | 11180                                                                                                           | 9514       |
| DATA DURATION (h)        | 41.92                   | 4.66                                                                                                            | 3.96       |
| PARTICIPANTS             | 180                     |                                                                                                                 |            |
| GESTURE TYPE             | positive samples        | double finger pinch                                                                                             |            |
|                          | negative samples        | including single finger pinch, single fist, double fist, typing, tapping on a table, and arms hanging naturally |            |

- **Large-Scale Pre-training Subset:** This subset focuses on recognizing a single, predefined gesture ("double finger pinch"). It contains nearly 50 hours of rich, labeled data from 180 participants performing positive and negative examples, supporting robust model pre-training. Details are in Table.2.
- **Few-Shot Custom Gesture Subset:** This subset targets personalized gesture recognition. It includes approximately 2 hours of high-quality data from 10 participants performing 9 predefined custom gestures (e.g., "wrist down flip," "flip coin") in authentic settings. This subset is designed explicitly to simulate and evaluate few-shot learning for new, user-defined gestures. Key statistics for the custom gesture subset are listed in the GRHand column of Table.1.

The GRHand dataset is uniquely valuable due to its foundation in real commercial device usage, combining large-scale pre-training data with a diverse, realistic custom gesture evaluation suite. It effectively supports the development and validation of algorithms for practical few-shot and personal-

ized gesture recognition.

**UMAHand Dataset:** We employ the UMAHand dataset, a public inertial sensor dataset released in 2024 for hand movement recognition. It contains data from 25 participants performing 29 categories of fine-grained hand activities (e.g., drinking, writing, using a phone) in a home environment. A Shimmer 3 IMU worn on the wrist simultaneously captured raw triaxial accelerometer and gyroscope signals alongside video. Its primary strengths for our work are its focus on realistic, fine-grained wrist motions and its provision of synchronized multimodal data, making it a high-quality benchmark for evaluating wrist-worn activity recognition algorithms. Key statistics are provided in the UMAHand column of Table.1.

### 3.2. Evaluation Metrics

To ensure a comprehensive and fair evaluation, we employ the following four standard classification metrics alongside a critical model efficiency indicator.

Classification Performance Metrics:

- **Accuracy:** The proportion of correctly classified samples over the total number of samples, offering an overall performance summary.
- **Precision:** The proportion of true positives among all samples predicted as positive, reflecting the model's exactness and ability to avoid false alarms.
- **Recall:** The proportion of true positives correctly identified among all actual positive samples, reflecting the model's completeness and ability to capture all relevant instances.
- **F1-Score:** The harmonic mean of precision and recall, providing a single balanced metric that is especially informative when class distribution is uneven.

Model Efficiency Metric:

Table 3. Comparison of F1-Score Statistical Results for Various Target Gestures in the UMAHand and GRHand Datasets. The red-highlighted column shows that our lightweight feature extractor (BiCroNet-DE) outperforms AppleDE and ImpTCNN-DE, while the blue-highlighted column demonstrates that our data augmentation methods (BiCroNet-DG and BiCroNet-EG) further improve performance, especially Embedding Generator.

| DATASET | GESTURE                  | TYPE       | DIFFICULTY | AppleDE | ImpTCNN-DE | BiCroNet-DE | BiCroNet-DG | BiCroNet-EG | BiCroNet-NOAUG |
|---------|--------------------------|------------|------------|---------|------------|-------------|-------------|-------------|----------------|
| UMAHand | clapping hands           | continuous | easy       | 86.61   | 85.87      | 95.10       | 98.47       | 98.51       | 66.74          |
|         | combing hair             | continuous | easy       | 84.61   | 84.55      | 93.09       | 95.19       | 98.21       | 53.47          |
|         | zipping zippers          | continuous | easy       | 95.04   | 82.56      | 95.79       | 98.21       | 98.72       | 65.63          |
|         | buttoning buttons        | continuous | easy       | 88.35   | 83.39      | 97.24       | 98.85       | 96.29       | 59.48          |
|         | screwing screws          | continuous | easy       | 84.68   | 80.62      | 91.29       | 99.04       | 99.09       | 62.13          |
|         | brushing teeth           | continuous | medium     | 84.21   | 83.95      | 87.75       | 95.00       | 97.86       | 20.28          |
|         | waving hands             | continuous | medium     | 94.68   | 81.86      | 87.22       | 95.24       | 98.25       | 54.17          |
| GRHand  | five-finger bend         | transient  | medium     | 85.70   | 79.34      | 91.19       | 93.20       | 96.93       | 56.01          |
|         | clapping                 | transient  | medium     | 90.05   | 82.37      | 91.05       | 90.09       | 98.98       | 53.59          |
|         | wrist down flip          | transient  | medium     | 83.08   | 84.52      | 88.60       | 93.57       | 98.64       | 53.68          |
|         | side wrist flip          | transient  | medium     | 92.29   | 84.77      | 92.82       | 94.24       | 97.27       | 53.56          |
|         | five finger snap         | transient  | hard       | 89.38   | 80.37      | 89.58       | 92.32       | 96.60       | 53.59          |
|         | peace sign with one fist | transient  | hard       | 86.75   | 77.56      | 87.18       | 89.94       | 97.27       | 55.39          |
|         | flip coin                | transient  | hard       | 88.94   | 75.78      | 89.55       | 85.68       | 94.11       | 55.08          |
|         | double wrist flip        | transient  | hard       | 82.33   | 86.04      | 86.67       | 89.18       | 96.92       | 55.46          |
|         | snapping                 | transient  | hard       | 88.57   | 78.57      | 88.99       | 90.72       | 88.49       | 54.23          |

- **Parameter Count:** The number of trainable parameters in the feature extraction network is counted. To ensure a fair comparison, the output feature dimension of all compared feature extractors is standardized, guaranteeing that the subsequent classifier shares an identical parameter scale across all methods. This metric directly reflects model complexity and is a key determinant of practical deployability on resource-constrained devices.

### 3.3. Experimental Setup

**Test Strategy:** To comprehensively and fairly evaluate the effectiveness and accuracy of the proposed algorithm, we conducted experiments on both the UMAHand and GRHand datasets.

**Hyperparameter Tuning:** All model hyperparameters, including learning rate, batch size, and network depth, were optimized via a grid search on a held-out validation set derived from the training subjects. This process ensures that each compared model achieves its peak potential under a fair and consistent tuning framework.

**Dataset Subsets:** From the UMAHand dataset, we selected a subset of 7 distinct gestures (e.g., brushing teeth, clapping, combing hair) that represent clearly recognizable daily activities. From the GRHand custom gesture subset, we selected all 9 gestures (e.g., five-finger bend, flip coin, wrist down flip), which were designed to prioritize natural and fluid hand interactions suitable for wrist-worn smart devices. This selection provides a balanced testbed covering both common activities (UMAHand) and novel, application-specific gestures (GRHand).

### 3.4. Analysis of Experimental Results

Table.3 shows the average F1-Score results and related information for various actions on the UMAHand and GR-

Table 4. Test Results of Seven Gestures on the UMAHand Dataset. Our proposed Delta Generator and Embedding Generator consistently outperform the baseline Delta Encoder. BiCroNet with Embedding Generator achieves the best performance with only 32,642 parameters, reducing model size by 27.8% compared to baseline methods.

| FEATURE EXTRACTOR |       | DATA AUG | UMAHand |       |       |       |
|-------------------|-------|----------|---------|-------|-------|-------|
| METHOD            | PARAM |          | A       | R     | F1    | P     |
| Apple             | 56752 | D-E      | 89.06   | 69.91 | 88.31 | 88.83 |
| Improved TCNN     | 45198 | D-E      | 82.18   | 81.61 | 83.26 | 61.47 |
|                   |       | D-G      | 97.39   | 92.88 | 97.36 | 99.93 |
|                   |       | E-G      | 98.10   | 94.94 | 98.08 | 99.98 |
| BiCroNet          | 32642 | NO-AUG   | 66.04   | 50.29 | 54.56 | 59.56 |
|                   |       | D-E      | 92.59   | 84.61 | 92.50 | 94.09 |
|                   |       | D-G      | 97.12   | 93.49 | 97.14 | 98.84 |
|                   |       | E-G      | 98.16   | 96.05 | 98.13 | 98.06 |
|                   |       |          |         |       |       |       |

Hand datasets. A comparison of the AppleDE, ImpTCNN-DE, and BiCroNet-DE columns demonstrates that our proposed, significantly lighter feature extractor has a clear accuracy advantage over the comparison methods (see red highlight). A comparison of the BiCroNet-DE, BiCroNet-DG, and BiCroNet-EG columns demonstrates that our two proposed data augmentation methods significantly outperform the comparison methods, especially the Embedding Generator, resulting in a significant improvement in classification performance (see blue highlight). Additionally, the BiCroNet-NOAUG column shows the performance of our model without any data augmentation. The substantial performance gap between BiCroNet-NOAUG and its augmented counterparts highlights the necessity of effective data augmentation. Furthermore, it is clear that our proposed custom gesture recognition algorithm can effectively handle various gestures, whether transient or continuous, easy or hard.

The complete experimental results are summarized in Ta-

Table 5. Test Results of Nine Gestures on the GRHand Dataset. Our data augmentation methods significantly improve accuracy across all feature extractors. BiCroNet with Embedding Generator achieves the best overall results, validating the effectiveness of our approach on diverse gesture types.

| FEATURE EXTRACTOR |       | DATA<br>AUG | GRHand       |              |              |              |
|-------------------|-------|-------------|--------------|--------------|--------------|--------------|
| METHOD            | PARAM |             | A            | R            | F1           | P            |
| Apple             | 56752 | D-E         | 88.37        | 68.20        | 87.45        | 94.72        |
| Improved TCNN     | 45198 | D-E         | 80.67        | 82.15        | 81.04        | 65.49        |
|                   |       | D-G         | 93.59        | 81.87        | 93.13        | 97.72        |
|                   |       | E-G         | 83.06        | 47.17        | 77.99        | 75.90        |
| BiCroNet          | 32642 | NO-AUG      | 67.58        | 51.03        | 54.51        | 58.27        |
|                   |       | D-E         | 89.55        | 86.77        | 89.51        | 82.34        |
|                   |       | D-G         | 91.06        | 87.39        | 90.99        | 85.44        |
|                   |       | E-G         | <b>96.30</b> | <b>89.11</b> | <b>96.13</b> | <b>99.16</b> |

ble.4 and Table.5, which report the per-gesture performance on the UMAHand and GRHand datasets, respectively. To facilitate comparison in cases where the top-performing methods are very close, the best result for each metric is highlighted in bold. Furthermore, to visualize the distribution and stability of model performance across all 16 selected gestures, we provide box plots for each evaluation metric in Fig.3.

Based on the accuracy metrics presented in the tables, the following key observations can be made:

- Whether using the Improved TCNN or our BiCrossNet as the feature extractor, the experimental groups employing our proposed Delta Generator and Embedding Generator consistently achieve significantly higher classification accuracy than the baseline Delta Encoder method. For instance, in the Improved TCNN control group on the UMAHand dataset, the Delta Generator and Embedding Generator improve accuracy by 18.51% and 19.37%, respectively, over the Delta Encoder baseline. This demonstrates a substantial advancement in data augmentation quality.
- When comparing models that use the same data augmentation method, our BiCrossNet Feature Extractor achieves a highly competitive performance-efficiency trade-off. For example, with the Embedding Generator, BiCrossNet reduces the number of model parameters by approximately 27.78% compared to the Improved TCNN extractor, while the average accuracy increases about 0.07%, remaining well above the practical usability threshold. This result validates the effectiveness of our dual-branch, cross-attention design in maintaining high accuracy with a markedly more compact model.
- The BiCroNet-NOAUG results show the critical role of data augmentation: without it, F1-Scores drop to

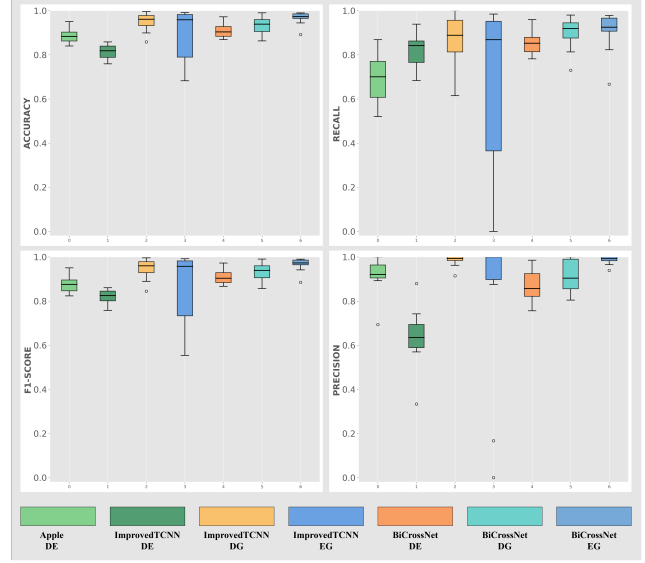


Figure 3. Distribution of Classification Metrics across Methods. The box plots show that our BiCrossNet Feature Extractor with Embedding Generator achieves both the highest performance and the smallest interquartile range, outperforming comparison methods with superior accuracy and robustness.

54.56% and 54.51% on UMAHand and GRHand, respectively, while the Embedding Generator raises them to 98.13% and 96.13%, which demonstrates that effective augmentation is essential for achieving competitive performance on limited data.

The box plots in Fig.3 provide a detailed view of performance distribution and robustness, leading to three key findings:

- The BiCrossNet Feature Extractor combined with the Embedding Generator data augmentation method achieves the highest median across all accuracy metrics. This indicates that in most gesture actions and usage scenarios, our proposed method significantly outperforms comparison methods such as AppleDE and Improved TCNN (which incorporates our Delta Generator or Embedding Generator as a data augmentation method).
- Similarly, considering the dispersion of the box plots, our proposed method has a significantly smaller interquartile range than other methods. This indicates that it exhibits less performance volatility across 16 different gesture action test scenarios, meaning its performance is less sensitive to scene changes and possesses stronger robustness.
- Furthermore, a comparison of the box plots for Improved TCNN and our proposed feature extractor combined with the three data augmentation methods clearly

shows that our feature extractor can better extract effective feature information from the augmented training data, thereby achieving significantly better classification accuracy and robustness.

Fig.4 shows the radar chart of the manifold similarity of the feature embedding sets corresponding to the amplified samples for nine different gestures in the GRHand dataset, as well as the bar chart of the overall average.

- We statistically analyzed the manifold similarity between the feature embedding sets generated by Delta Encoder, Delta Generator, and Embedding Generator and the feature embedding sets of the actual collected samples in the feature embedding space. A higher manifold similarity indicates that the generated samples better maintain the manifold of the original data, meaning the amplified training data samples are closer to the actual collected data. This is the essential requirement of our data augmentation methods.
- It is clearly evident that the Delta Generator and Embedding Generator proposed in this paper achieve significantly more ideal data sample augmentation compared to Apple’s Delta Encoder.

Table 6. Comparison of Inference Performance on an Intel Xeon Gold 6342 CPU @ 2.80GHz.

| Model (Params)                               | RAM (KiB)     | Latency (ms) |
|----------------------------------------------|---------------|--------------|
| BiCrossNet (32,642)<br>+ Custom Head (2,210) | <b>260.18</b> | <b>9.16</b>  |
| Apple (56,752)<br>+ Custom Head (4,002)      | 359.36        | 11.07        |
| TCNN (45,198)<br>+ Custom Head (1,762)       | 282.78        | 10.89        |

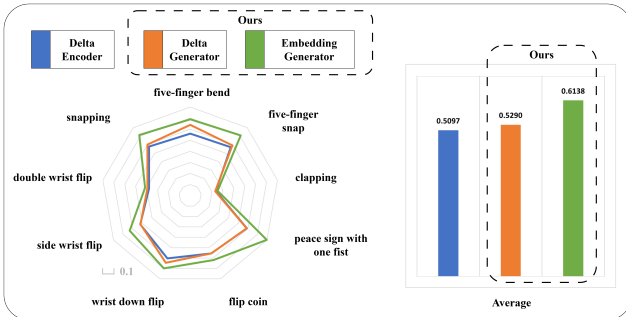


Figure 4. The Radar Chart of the Manifold Similarity of Samples Generated by Three Data Augmentation Methods across Nine Custom Gestures. Higher values indicate better preservation of the original data manifold. Our proposed Delta Generator and Embedding Generator consistently outperform Delta Encoder.

Table 7. Inference Memory Footprint and Dependencies of Three Data Augmentation Methods on an Intel Xeon Gold 6342 CPU @ 2.80GHz.

| Method              | RAM (MiB) | Inference Dependencies                                |
|---------------------|-----------|-------------------------------------------------------|
| Delta Encoder       | 4.16      | Pre-computed Delta Vectors<br>+ Encoder-Decoder Model |
| Delta Generator     | 1.11      | Encoder-Decoder Model<br>+ Delta Generator Model      |
| Embedding Generator | 3.60      | Embedding Generator Model                             |

Table.6 and Table.7 report the inference efficiency of different feature extractors and data augmentation methods, respectively, measured on an Intel Xeon Gold 6342 CPU @ 2.80GHz. As shown in Table.6, our BiCrossNet achieves the lowest memory footprint (260.18 KiB) and fastest inference latency (9.16 ms) among all compared models, with 27.6% less RAM and 17.3% lower latency than the Apple model. Furthermore, Table.7 reveals that among the three data augmentation approaches, the Delta Generator is the most memory-efficient at only 1.11 MiB, while the Delta Encoder requires 4.16 MiB due to its dependency on pre-computed Delta vectors. The Embedding Generator, though slightly larger at 3.60 MiB, has the simplest inference dependency, requiring only the generator model itself without any pre-computed vectors.

## 4. Conclusions

This paper addresses the core pain points in the commercialization of custom gesture recognition technology by proposing an IMU-based BiCrossNet framework and a decoupled dual-generator data augmentation model. This achieves a high-efficiency balance between accuracy and model size in small-sample scenarios, fulfilling the research objectives and providing crucial support for the engineering deployment of the technology.

The research overcomes existing limitations through three core innovations: BiCrossNet’s dual-branch feature extraction and symmetrical cross-attention module, which simplifies the number of parameters, avoids computational redundancy, and enhances feature discriminative power and interpretability; the two decoupled generators decouple data augmentation memory usage from sample size, eliminating the need for additional storage of difference vector sets during inference, significantly saving memory; and the constructed dataset exceeding 40 hours (including the self-developed GRHand dataset) provides a reliable foundation for model validation.

Experiments show that the proposed framework’s average F1 score performance on the public UMAHand dataset is 98.13%, and 96.13% on the GRHand dataset. This

study constructs a practical few-sample gesture recognition scheme, advancing the practical application of IMU gesture recognition technology and laying the foundation for the deployment of gesture recognition engines in smart terminals. Future development can expand its multimodal adaptation capabilities, explore applications in complex scenarios, and facilitate the deep commercialization of the technology.

## Impact Statement

This paper presents work aimed at advancing the field of Machine Learning, with a specific focus on End-side sensing for smart watch/bracelet scenarios. In terms of ethical aspects, all experimental data used in our research are either publicly available benchmark datasets or anonymized private data with informed consent from participants, strictly complying with data privacy and protection regulations. Our work may bring potential societal consequences such as improving the efficiency of human-computer interaction and optimizing the user experience of intelligent devices; none of these consequences involve ethical risks that require specific highlighting here.

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## References

- Choi, J.-H., Choi, H.-T., Kim, K.-T., Jung, J.-S., Lee, S.-H., and Chang, W.-D. Gesture classification using a smartwatch: Focusing on unseen non-target gestures. *Applied Sciences*, 15(9):4867, 2025.
- Dwork, C., McSherry, F., Nissim, K., and Smith, A. Calibrating noise to sensitivity in private data analysis. In *Theory of cryptography conference*, pp. 265–284. Springer, 2006.
- Gomes, P. R. I., Castro, M. S. d., and Nascimento, T. H. Gesture recognition methods using sensors integrated into smartwatches: Results of a systematic literature review. In *Proceedings of the XXII Brazilian Symposium on Human Factors in Computing Systems*, pp. 1–11, 2023.
- Kim, J., Lee, J., and Kim, W. Transferable convolutional neural networks for imu-based motion gesture recognition in human-machine interaction. In *2024 24th International Conference on Control, Automation and Systems (ICCAS)*, pp. 61–66. IEEE, 2024.
- Langley, P. Crafting papers on machine learning. In Langley, P. (ed.), *Proceedings of the 17th International Conference on Machine Learning (ICML 2000)*, pp. 1207–1216, Stanford, CA, 2000. Morgan Kaufmann.
- Laplace, P. S. Mémoire sur la probabilité de causes par les événements. *Mémoire de l’académie royale des sciences*, 1774.
- Li, D., Kang, P., Zhu, K., Li, J., and Shull, P. B. Feasibility of wearable ppg for simultaneous hand gesture and force level classification. *IEEE Sensors Journal*, 23(6):6008–6017, 2023.
- Shahi, S., Mollyn, V., Tymoszek Park, C., Kang, R., Liberman, A., Levy, O., Gong, J., Bedri, A., and Laput, G. Vision-based hand gesture customization from a single demonstration. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–14, 2024.
- Xu, C., Pathak, P. H., and Mohapatra, P. Finger-writing with smartwatch: A case for finger and hand gesture recognition using smartwatch. In *Proceedings of the 16th international workshop on mobile computing systems and applications*, pp. 9–14, 2015.
- Xu, X., Gong, J., Brum, C., Liang, L., Suh, B., Gupta, S. K., Agarwal, Y., Lindsey, L., Kang, R., Shahsavari, B., et al. Enabling hand gesture customization on wrist-worn devices. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pp. 1–19, 2022.